

NGÂN HÀNG CÂU HỎI THI TỰ LUẬN

Tên học phần: PHÂN TÍCH THIẾT KẾ HỆ THỐNG THÔNG TIN
(INFORMATION SYSTEM ANALYSIS AND DESIGN)
Mã học phần: INT1342_CLC
Ngành đào tạo: CÔNG NGHỆ THÔNG TIN CHẤT LƯỢNG CAO
Trình độ đào tạo: ĐẠI HỌC CHÍNH QUY

Cấu trúc đề

Được viết dưới dạng các đề thi trọn vẹn trong đó yêu cầu trả lời các câu hỏi hay thực hiện các yêu cầu ứng với một hệ thống được mô tả trong đề bài. Mỗi đề gồm 5 câu, mỗi câu 2 điểm.

Thời gian làm bài: 90 phút

Ngân hàng câu hỏi thi này đã được thông qua bộ môn và nhóm cán bộ giảng dạy học phần.

Hà Nội, ngày tháng năm 2022

Trưởng khoa

Trưởng bộ môn

Giảng viên chủ trì hiệu chỉnh

Nguyễn Duy Phương

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Table of content

Table of content.....	2
Subject No. 01.....	3
Subject No. 02.....	4
Subject No. 03.....	5
Subject No. 04.....	6
Subject No. 05.....	7
Subject No. 06.....	8
Subject No. 07.....	9
Subject No. 08.....	10
Subject No. 09.....	11
Subject No. 10.....	12
Subject No. 11.....	13
Subject No. 12.....	14
Subject No. 13.....	15
Subject No. 14.....	16
Subject No. 15.....	17
Subject No. 16.....	18
Subject No. 17.....	19
Subject No. 18.....	20
Subject No. 19.....	21
Subject No. 20.....	22
Subject No. 21.....	23
Subject No. 22.....	24
Subject No. 23.....	25
Subject No. 24.....	26
Subject No. 25.....	27
Subject No. 26.....	28
Subject No. 27.....	29
Subject No. 28.....	30
Subject No. 29.....	31
Subject No. 30.....	32
Subject No. 31.....	33
Subject No. 32.....	34
Subject No. 33.....	35
Subject No. 34.....	36
Subject No. 35.....	37
Subject No. 36.....	38
Subject No. 37.....	39

Subject No. 01

Duration: 90 minutes

A Library management system of a University (**LibMan**) allows to manage the common documents (books, textbooks, journals, etc.), manage borrowing and returning activities and status performed by students. The system allows:

- *Management staff*: view statistical reports of status of documents and readers via loan as well as imported documents from suppliers.
- *Library staff*: manage documents, readers, suppliers (add, delete, change), search, lend documents, receive and return documents from readers, import documents from suppliers.
- *Readers*: borrow documents, return documents directly to staff, search for document information, register for online reader cards.

Considering two modules and answering questions:

- *Reader finds document information*: selects the document search menu → enters the document's name to search → the system displays a list of documents whose name contains the entered keyword → clicks on a document to view details → the system shows detailed information about the document.
- *Library staff lends documents to readers*: selects the document loan menu → scan reader card (or search by code) → Repeats the following steps for all borrowed documents: scan document code (or search by code) → repeats until all borrowed materials are finished, then submit → prints the loan slip and delivers it to the reader.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 02

Duration: 90 minutes

A Library management system of a University (**LibMan**) allows to manage the common documents (books, textbooks, journals, etc.), manage borrowing and returning activities and status performed by students. The system allows:

- *Management staff*: view statistical reports of status of documents and readers via loan as well as imported documents from suppliers.
- *Library staff*: manage documents, readers, suppliers (add, delete, change), search, lend documents, receive and return documents from readers, import documents from suppliers.
- *Readers*: borrow documents, return documents directly to staff, search for document information, register for online reader cards.

Considering two modules and answering questions:

- *Reader registers as a reader card*: selects the menu to register as a reader card → enters the information of the reader and the card, clicks save → the system saves to database.
- *Reader returns documents*: selects the menu to receive and return documents → scans the reader's card (or search by code) → selects the documents that the reader brings to return in the list of borrowed documents (additional money fines for damage), late payment (if any) → shows returned invoices → confirms → prints payment slips and deliver to the reader (pay fines if any).

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 03

Duration: 90 minutes

A Library management system of a University (**LibMan**) allows to manage the common documents (books, textbooks, journals, etc.), manage borrowing and returning activities and status performed by students. The system allows:

- *Management staff*: view statistical reports of status of documents and readers via loan as well as imported documents from suppliers.
- *Library staff*: manage documents, readers, suppliers (add, delete, change), search, lend documents, receive and return documents from readers, import documents from suppliers.
- *Readers*: borrow documents, return documents directly to staff, search for document information, register for online reader cards.

Considering two modules and answering questions:

- *Reader finds document information*: selects the document search menu → enters the document's name to search → the system displays a list of documents whose name contains the entered keyword → clicks on a document to view details → the system shows detailed information about the document.
- *Library staff imports documents from suppliers*: selects the document import menu → finds suppliers by name (add new if not available) → Repeats the following steps for all imported documents: scans the document code (or search by code, add new if not available) + enters quantity and unit price → repeats until all imported documents are finished, submits → prints invoice and pays to the supplier.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 04

Duration: 90 minutes

A Library management system of a University (**LibMan**) allows to manage the common documents (books, textbooks, journals, etc.), manage borrowing and returning activities and status performed by students. The system allows:

- *Management staff*: view statistical reports of status of documents and readers via loan as well as imported documents from suppliers.
- *Library staff*: manage documents, readers, suppliers (add, delete, change), search, lend documents, receive and return documents from readers, import documents from suppliers.
- *Readers*: borrow documents, return documents directly to staff, search for document information, register for online reader cards.

Considering two modules and answering questions:

- *Reader registers as a reader card*: selects the menu to register as a reader card → enters the information of the reader and the card, clicks save → the system saves to database.
- *Management staff views document statistics*: selects the menu to view statistical reports → selects document statistics by loan amount → selects the start and end date of statistics → view document statistics → clicks on a document → views details of borrowed documents → clicks a loan → view details of the corresponding loan slip.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 05

Duration: 90 minutes

A Library management system of a University (**LibMan**) allows to manage the common documents (books, textbooks, journals, etc.), manage borrowing and returning activities and status performed by students. The system allows:

- *Management staff*: view statistical reports of status of documents and readers via loan as well as imported documents from suppliers.
- *Library staff*: manage documents, readers, suppliers (add, delete, change), search, lend documents, receive and return documents from readers, import documents from suppliers.
- *Readers*: borrow documents, return documents directly to staff, search for document information, register for online reader cards.

Considering two modules and answering questions:

- *Reader finds document information*: selects the document search menu → enters the document's name to search → the system displays a list of documents whose name contains the entered keyword → clicks on a document to view details → the system shows detailed information about the document.
- *Management staff views reader statistics*: selects the menu to view the statistical reports → chooses to view the reader statistics by loan number → selects the start and end date of the statistics → views the reader statistics → clicks on a reader → views the details of the times the reader borrows → clicks a loan → views the details of the corresponding loan slip.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 06

Duration: 90 minutes

A Library management system of a University (**LibMan**) allows to manage the common documents (books, textbooks, journals, etc.), manage borrowing and returning activities and status performed by students. The system allows:

- *Management staff*: view statistical reports of status of documents and readers via loan as well as imported documents from suppliers.
- *Library staff*: manage documents, readers, suppliers (add, delete, change), search, lend documents, receive and return documents from readers, import documents from suppliers.
- *Readers*: borrow documents, return documents directly to staff, search for document information, register for online reader cards.

Considering two modules and answering questions:

- *Reader registers as a reader card*: selects the menu to register as a reader card → enters the information of the reader and the card, clicks save → the system saves to database.
- *Management staff views supplier statistics*: selects the menu to view statistical reports → selects to view supplier statistics by imported volume → selects the start and end date of statistics → views supplier statistics → clicks on a supplier → views list of imported invoices from the supplier → clicks an invoice → views details of the corresponding imported invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 07

Duration: 90 minutes

An online supermarket management system enables management staff, warehouse staff, saler and customer to use:

- Management staff: view the types of statistics: items, suppliers and revenue.
- Warehouse staff: import goods from suppliers, manage (add, delete, change) item information and supplier information, browse online orders and export to delivery staff.
- Saler: selling goods at the counter to customers
- Customer: register as a member, search, order online, buy directly at the counter

Considering two modules and answering questions:

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Warehouse staff imports goods from suppliers:** selects the import menu → find suppliers by name (if not, add a new one) → Repeats the following steps for all imported goods: selects search for items by name (if not already available, then add new) → enters quantity + unit price → repeats until all the imported goods are finished, then submits → prints out the invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 08

Duration: 90 minutes

An online supermarket management system enables management staff, warehouse staff, saler and customer to use:

- Management staff: view the types of statistics: items, suppliers and revenue.
- Warehouse staff: import goods from suppliers, manage (add, delete, change) item information and supplier information, browse online orders and export to delivery staff.
- Saler: selling goods at the counter to customers
- Customer: register as a member, search, order online, buy directly at the counter

Considering two modules and answering questions:

- **Warehouse staff edits item information:** selects the item management menu → selects the function to edit item information → finds items by name to edit → enters item information and clicks edit → the system saves information to the database.
- **Customer orders online:** chooses to order → Repeats the following steps until the end of ordered item: chooses to find items by name → enters quantity → repeats until all items are out, then goes to cart submit → saves order.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 09

Duration: 90 minutes

An online supermarket management system enables management staff, warehouse staff, saler and customer to use:

- Management staff: view the types of statistics: items, suppliers and revenue.
- Warehouse staff: import goods from suppliers, manage (add, delete, change) item information and supplier information, browse online orders and export to delivery staff.
- Saler: selling goods at the counter to customers
- Customer: register as a member, search, order online, buy directly at the counter

Considering two modules and answering questions:

- **Customer searches for items:** selects the item search menu → enters the name of the item to search → the system displays a list of items whose name contains the entered keyword → clicks on an item to view details → the system displays detailed information about the item.
- **Warehouse staff approves orders + export:** selects menu to browse orders → selects unexported orders in the list → selects delivery staff and update the status of exported order → prints invoices and deliver goods + invoices for the delivery staff .

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 10

Duration: 90 minutes

An online supermarket management system enables management staff, warehouse staff, saler and customer to use:

- Management staff: view the types of statistics: items, suppliers and revenue.
- Warehouse staff: import goods from suppliers, manage (add, delete, change) item information and supplier information, browse online orders and export to delivery staff.
- Saler: selling goods at the counter to customers
- Customer: register as a member, search, order online, buy directly at the counter

Considering two modules and answering questions:

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Management staff views item statistics:** selects the menu to view statistical reports → selects view item statistics by revenue → selects start and end dates → Views item statistics → click on an item view details → views related transaction list.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 11

Duration: 90 minutes

An online supermarket management system enables management staff, warehouse staff, saler and customer to use:

- Management staff: view the types of statistics: items, suppliers and revenue.
- Warehouse staff: import goods from suppliers, manage (add, delete, change) item information and supplier information, browse online orders and export to delivery staff.
- Saler: selling goods at the counter to customers
- Customer: register as a member, search, order online, buy directly at the counter

Considering two modules and answering questions:

- **Customer searches for items:** selects the item search menu → enters the name of the item to search → the system displays a list of items whose name contains the entered keyword → clicks on an item to view details → the system displays detailed information about the item.
- **Management staff views supplier statistics:** selects the menu to view statistical reports → selects to view supplier statistics by revenue → selects start and end dates → Views supplier statistics → clicks on a supplier to view details → views statistics of supplier's imported invoice → click view an imported invoice → view the selected invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 12

Duration: 90 minutes

An online supermarket management system enables management staff, warehouse staff, saler and customer to use:

- Management staff: view the types of statistics: items, suppliers and revenue.
- Warehouse staff: import goods from suppliers, manage (add, delete, change) item information and supplier information, browse online orders and export to delivery staff.
- Saler: selling goods at the counter to customers
- Customer: register as a member, search, order online, buy directly at the counter

Considering two modules and answering questions:

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Management staff views customer statistics:** selects the menu to view statistical reports → chooses to view customer statistics by revenue → selects the start and end date → Views statistics of customers → clicks on a customer to view details → views list of related transactions.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 13

Duration: 90 minutes

A cinema management system could be used by management staff, saler and customers:

- Management staff: view the types of statistics: movies, customers and revenue. Schedule showtimes, manage movie and theater information (add, edit, delete).
- Saler: selling tickets at the counter to customers, issuing membership cards to customers
- Customer: register for membership, search, buy tickets online, buy tickets at the counter

Considering two modules and answering questions

- **Management staff adds movie information:** selects the movie information management menu → selects the function to add movie information → enters movie information and clicks add → the system saves to the database.
- **Customer purchases online ticket:** selects the ticket purchase menu → selects a movie from the list → selects a viewing time → selects a seat → Enters the loyalty card code (if any) and confirms ticket purchase → pays → prints system tickets and invoices.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 14

Duration: 90 minutes

A cinema management system could be used by management staff, saler and customers:

- Management staff: view the types of statistics: movies, customers and revenue. Schedule showtimes, manage movie and theater information (add, edit, delete).
- Saler: selling tickets at the counter to customers, issuing membership cards to customers
- Customer: register for membership, search, buy tickets online, buy tickets at the counter

Considering two modules and answering questions

- *Customer registers as a member*: chooses to register as a member → enters personal information and clicks register → the system saves information to the database.
- *Saler sales ticket for customers* (at the counter): selects the ticketing menu → asks for the movie name and searches for the movie → selects the movie from the list of results → selects the viewing time → selects the seat → Enters the loyalty card code (if any) and confirms ticket purchase → payment → prints ticket and invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 15

Duration: 90 minutes

A cinema management system could be used by management staff, saler and customers:

- Management staff: view the types of statistics: movies, customers and revenue. Schedule showtimes, manage movie and theater information (add, edit, delete).
- Saler: selling tickets at the counter to customers, issuing membership cards to customers
- Customer: register for membership, search, buy tickets online, buy tickets at the counter

Considering two modules and answering questions

- **Customer searches for movie information:** selects the menu to find movie information → enters the movie name to find it → the system displays a list of movies whose name contain the entered keywords → clicks on a movie to view details → the system displays information movie details.
- **Showtimes scheduling:** selects the showtime management menu → selects add more showtime → selects the time slot and showtime date → finds and selects the available screening room at the selected time slot → selects the movie from the showing list (add new if not) → confirms → the system saves the schedule.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 16

Duration: 90 minutes

A cinema management system could be used by management staff, saler and customers:

- Management staff: view the types of statistics: movies, customers and revenue. Schedule showtimes, manage movie and theater information (add, edit, delete).
- Saler: selling tickets at the counter to customers, issuing membership cards to customers
- Customer: register for membership, search, buy tickets online, buy tickets at the counter

Considering two modules and answering questions

- *Customer registers as a member*: chooses to register as a member → enters personal information and clicks register → the system saves information to the database.
- *Management staff views movie statistics*: selects the menu to view statistical reports → chooses to view movie statistics by revenue → selects start and end date → Views movie statistics → clicks on a movie to view details → views statistics of movie showtimes → clicks on a showtime → views statistics of ticket sales of the showtime.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 17

Duration: 90 minutes

A cinema management system could be used by management staff, saler and customers:

- Management staff: view the types of statistics: movies, customers and revenue. Schedule showtimes, manage movie and theater information (add, edit, delete).
- Saler: selling tickets at the counter to customers, issuing membership cards to customers
- Customer: register for membership, search, buy tickets online, buy tickets at the counter

Considering two modules and answering questions

- **Customer searches for movie information:** selects the menu to find movie information → enters the movie name to find it → the system displays a list of movies whose name contain the entered keywords → clicks on a movie to view details → the system displays information movie details.
- **Management staff views customer statistics:** selects the menu to view statistical reports → selects view customer statistics by revenue → selects start date, end date → Views customer statistics → clicks on a customer to view details → views statistics of transactions of the selected customer.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 18

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- *Management staff adds new dish*: selects the menu to manage dish information → selects the function of adding dish information → enters dish information and clicks add → The system saves to the database.
- *Customer books tables online*: selects the booking menu → selects a date and time to find an available table → selects a table from the list of results → enters the customer's personal information and confirms → The system saves to the database.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 19

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- **Management staff edits dish information:** selects the menu to manage dish information → selects the function to edit dish information → searches for dishes by name to edit → selects dishes from the list of results to edit → enters information dishes and clicks save → saves to the database.
- **Customer orders online:** chooses the menu to order → finds the booked table by the customer's name (phone number) → chooses his/her own table → Repeats until the end of the ordered dishes: searches for dishes by name/type → selects the dish from the list of results → enters the quantity → at the end of the ordered list, confirms → the system saves to the database.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 20

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- **Customer searches for dish information:** selects the menu to find dish information → enters the name of the dish to search → the system displays a list of dishes with the name of the keyword entered → clicks on a dish to view details → the system displays detailed information about the dish.
- **Sale staff receives payment:** selects the payment menu → finds the table by name → selects the correct table in the list of results → an invoice will appear containing the list of ordered dishes, unit price, total amount, total amount → confirms receive → prints invoices delivered to customers and receive money.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 21

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- **Management staff adds new dish:** selects the menu to manage dish information → selects the function of adding dish information → enters dish information and clicks add → The system saves to the database.
- **Warehouse staff imports ingredients:** selects the menu to import ingredients → finds suppliers by name (add new if not exist) → Repeats until the end of import ingredients: searches for ingredients by name (add new if not exist) → selects the ingredients from the result list → enters the quantity → at the end of the list of imported ingredients, confirms → prints the invoice and pays to the supplier.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 22

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- **Management staff edits dish information:** selects the menu to manage dish information → selects the function to edit dish information → searches for dishes by name to edit → selects dishes from the list of results to edit → enters information dishes and clicks save → saves to the database.
- **Management staff sets up a combo menu:** selects combo management menu → chooses add new combo → Repeats until the end of all dishes wanted to put in combo: searches for dishes by name/category → select dishes from the list of results to put in combos → at the end of the list of dishes, confirms → the system saves to the database.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 23

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- **Customer searches for dish information:** selects the menu to find dish information → enters the name of the dish to search → the system displays a list of dishes with the name of the keyword entered → clicks on a dish to view details → the system displays detailed information about the dish.
- **Management staff views statistics of dishes:** selects menu to view statistical reports → selects dish statistics by revenue → chooses time to start and end statistics → views dish statistics → selects a dish to view details → views the list of order including the dish → selects view 1 order → views the selected order.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 24

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- *Management staff adds new dish*: selects the menu to manage dish information → selects the function of adding dish information → enters dish information and clicks add → The system saves to the database.
- Management staff views customer statistics: selects the menu to view statistical reports → selects customer statistics by revenue → selects the time to start and end statistics → views customer statistics → selects a customer to view expenses details → views the order list that the customer has called → selects view 1 order → views detail the selected order.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 25

Duration: 90 minutes

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- **Management staff edits dish information:** selects the menu to manage dish information → selects the function to edit dish information → searches for dishes by name to edit → selects dishes from the list of results to edit → enters information dishes and clicks save → saves to the database.
- **Management staff view supplier statistics:** selects the menu to view the statistical report → selects the supplier statistics according to the imported quantity → selects the start and end time of statistics → views supplier statistics → selects a supplier view details → views list of imported invoice of the supplier → selects view 1 imported invoice → views the detail of selected imported invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 26

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- **Customer serches for service information:** selects the menu to find service / spare parts information → enters the service / part name to search → the system displays a list of services / parts whose name contains the entered keyword → clicks on a service/part to view details → the system displays detailed information about the service/part.
- **Warehouse staff imports parts:** selects the menu to import parts → finds and selects suppliers by name (add new if not exist) → Repeats until the end of part list wanted to import: finds and selects spare parts by name (add new if not exist) → enters quantity, unit price → at the end of part list, confirms → prints import invoice and pays to supplier.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 27

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- ***Customer books an appointment online:*** selects the appointment booking menu → chooses a date and time → enters the customer's personal information and confirms → the system saves to the database.
- ***Sale staff receives the car in for repair:*** selects the menu to receive the car → finds and selects the customer by name (add a new one if not exist) → Repeats until the end of service and spare parts list to use: finds and selects the spare part/service by name → at the end of the list to use, selects service technician → confirms → saves to the database.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 28

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- **Customer serches for service information:** selects the menu to find service / spare parts information → enters the service / part name to search → the system displays a list of services / parts whose name contains the entered keyword → clicks on a service/part to view details → the system displays detailed information about the service/part.
- **Sale staff receives payment:** selects the payment menu → finds and selects customers by name → view detailed invoices containing a list of used spare parts and services (can be added if needed) → invoice confirms → Prints invoices and receive payments from customers.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 29

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- **Customer books an appointment online:** selects the appointment booking menu → chooses a date and time → enters the customer's personal information and confirms → the system saves to the database.
- **Management staff views service/parts statistics:** selects the menu to view statistical reports → selects service/parts statistics by revenue → selects start and end date of statistics → views service/parts statistics → clicks a service/part to view details → views a list of invoices including the service/part → clicks to view an invoice → views the detail of selected invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 30

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- **Customer serches for service information:** selects the menu to find service / spare parts information → enters the service / part name to search → the system displays a list of services / parts whose name contains the entered keyword → clicks on a service/part to view details → the system displays detailed information about the service/part.
- **Management staff views customer statistics:** selects the menu to view statistical reports → selects customer statistics by revenue → selects the start and end date of statistics → views customer statistics → clicks on a customer → view details of the invoices of the customer → clicks on an invoice → view the detail selected invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 31

Duration: 90 minutes

A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- **Customer books an appointment online:** selects the appointment booking menu → chooses a date and time → enters the customer's personal information and confirms → the system saves to the database.
- **Management staff views statistics of supplier:** selects the menu to view the statistical reports → selects the supplier statistics by imported quantity → selects the start and end date of the statistics → views the supplier statistics → clicks on a supplier → view the details of the supplier's import invoices → clicks on an invoice → views the selected imported invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 32

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

Considering two modules and answering questions

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Management staff imports goods from suppliers:** selects the import menu → find suppliers by name (if not, add a new one) → Repeats the following steps for all imported goods: selects search for items by name (if not already available, then add new) → enters quantity + unit price → repeats until all the imported goods are finished, then submits → prints out the invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 33

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

Considering two modules and answering questions

- **Customer searches for items:** selects the item search menu → enters the name of the item to search → the system displays a list of items whose name contains the entered keyword → clicks on an item to view details → the system displays detailed information about the item.
- **Customer orders online:** chooses to order → Repeats the following steps until the end of ordered item: chooses to find items by name → enters quantity → repeats until all items are out, then goes to cart submit → saves order.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 34

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

Considering two modules and answering questions

- **Customer tracks invoice:** selects the invoice tracking menu → clicks on the invoice from the list of ordered invoices → the system displays detailed information and current status of the invoice.
- **Sale staff approves orders + export:** selects menu to browse orders → selects unexported orders in the list → selects delivery staff and update the status of exported order → prints invoices and deliver goods + invoices for the delivery staff .

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 35

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

Considering two modules and answering questions

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Management staff views item statistics:** selects the menu to view statistical reports → selects view item statistics by revenue → selects start and end dates → Views item statistics → click on an item view details → views related transaction list.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 36

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

Considering two modules and answering questions

- **Customer searches for items:** selects the item search menu → enters the name of the item to search → the system displays a list of items whose name contains the entered keyword → clicks on an item to view details → the system displays detailed information about the item.
- **Management staff views customer statistics:** selects the menu to view statistical reports → chooses to view customer statistics by revenue → selects the start and end date → Views statistics of customers → clicks on a customer to view details → views list of related transactions.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

Subject No. 37

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

Considering two modules and answering questions

- **Customer tracks invoice:** selects the invoice tracking menu → clicks on the invoice from the list of ordered invoices → the system displays detailed information and current status of the invoice.
- **Management staff views supplier statistics:** selects the menu to view statistical reports → selects to view supplier statistics by revenue → selects start and end dates → Views supplier statistics → clicks on a supplier to view details → views statistics of supplier's imported invoice → click view an imported invoice → view the selected invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.